

The Ephemeris

Quarterly Edition January - March 2015

Volume 26 Number 01 - The Official Publication of the San Jose Astronomical Association



President's Message

From Rob Jaworski

At the most recent SJAA board meeting, the membership chair reported the club reaching the level of 350 members. This is an all-time high, at least during my board involvement since 2009. I attribute this milestone to the quality of the people that the club attracts. The SJAA is a very active educational nonprofit organization that takes to heart the mission to bring astronomy and science to everyone, and to make it as accessible as possible. This is evident in the wide variety of programs that the SJAA offers, most of which are available to anyone, free of charge and regardless of membership status. From professionals who come and give talks and the astronomy classes, to the very popular In Town Star Parties, and the School Star Party Program, not to mention helping get people involved in the hobby of astronomy via the QuickSTART program, the SJAA provides a valuable service, an outlet to people who may have a fledgling interest in astronomy, but don't know where or how to get started. The SJAA is there for them.

All these programs certainly don't run themselves, and that's where dedicated volunteers come into the picture. We have an entire spectrum of people contributing their time. There are those who come for the programs themselves, to take advantage of what's offered. Some of those people will come back to participate in those programs, such as bringing their telescope to Hogue Park to show the rings of Saturn to neighborhood kids for the first time. Some of those people will find enough satisfaction in that such that they will manage the program, for example, opening and closing the In Town Star Parties. From there, a few dedicated souls see the value in what they are doing to the extent that they get involved with more programs, and even serve on the board of directors. And a few of those may even step up to an officer position, such as Secretary, Treasurer and President.

I'd like to think that we do a decent job of cultivating this progression of involvement. But as usual, there is room for improvement. For some roles, we approach individuals directly to inquire about the possibility of getting involved. I routinely put out a call to the general membership to encourage people to consider contributing the most precious resource of all, time. There are plenty of ideas about innovative new programs that the SJAA could offer, but that precious resource is scarce unless there are others who could step up and run a program. It's also a good idea to rotate people through different roles to avoid burnout, which many organizations, including ours, have seen. This, in my mind, is to be avoided at all costs! And that's where you come in. If you'd like to get your feet wet and see what it's like to experience the satisfaction of running an educational program in a nonprofit setting, there are no better ways than to give it a try with the SJAA. This is a very welcoming group, we are all volunteers, and you could design and structure the program as you see fit. There are few rules, mostly just guidelines. Please consider it, and come to the next board meeting or event, or just seek me out to talk about your ideas. Or you can also follow the below link to the SJAA web page for a list of volunteer roles the SJAA needs help on at the Hogue Park events.

<http://www.sjaa.net/volunteer-at-hogue-park>

Speaking of avoiding burnout, the SJAA bylaws explicitly state that the President role cannot be held by the same person for more than two consecutive years. My term as President reaches that limit in March 2015. Therefore, I will be stepping aside for the next leader, whoever that may be. The board of directors meeting on March 7, 2015, starting at 6pm in the Hall at Hogue Park, will see an officers election from which will emerge our new President. It's open to all, please come and observe, or better yet, participate in the discussion!

Also note that we have a vacancy on the board. If you are interested in being part of the running of the SJAA, please come to the next board meeting and make yourself available. You'll be surprised at how interesting, fun and rewarding it actually is!

Editor's Note: From Sandy Mohan and Tom Piller

There are a number of contributors who devote large amounts of their time to make the SJAA club special and an asset to the star-gazing community in South Bay. With Rob Jaworski passing on the Presidential baton this month we thought this would be a good time to recognize his contribution to the SJAA cause. We would like to express our sincere appreciation to Rob for his tireless efforts in increasing SJAA membership and helping to make the Club so vibrant. We are also hoping that we don't lose our editorial buddy as Rob has been a big part of helping us bring the Ephemeris to life over the past year.

Thanks Rob!

March-May 2015 Events

Board & General Meetings

Saturday: 3/7, 4/4, 5/2, 5/30

(No General meeting on 3/7)

Board Meetings: 6 - 7:30pm

General Meetings: 7:30-9:30pm

Annual Auction

Sunday: 3/8 @10am

Solar Observing & Fix-It

Sunday: 3/1, 4/5, 5/3, 6/7

Solar observing: 2-4pm

Fix-It Day: 2-4pm

Hogue Star Party

Friday: 3/13, 3/27, 4/10, 5/8, 5/22

Intro to the Night Sky Class

In-Town Star party (Hogue)

Quick START (by appointment)

Friday: 3/6, 4/11, 5/9, 6/13

RCDO

Saturday 3/14, 4/11, 5/9, 6/6

Starry Nights Star Party (RCDO)

Unless noted above, please refer to the SJAA Web page for specific event times.

SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.

INSIDE THIS ISSUE

SJAA Annual Meeting.....	2-3
Member Observation Posts.....	4-6
Solar Observing.....	7
Ceres - Voyager Images.....	8-9
Member Astro images.....	10-12
DSSH.....	13
Kid Spot & Constellations.....	14
Club Updates.....	15-16
Membership form.....	17

SJAA Annual Meeting and Potluck

From Michael Packer



This year at the February 7th annual meeting, elections were held, food served, and a few members were recognized for making SJAA shine. Greg Claytor did an outstanding job of noting all that our clubs does and his son took the above picture of SJAA members in attendance. The Elected Board members for open seats in 2015 were Teruo Utsumi, Ed Wong, Lee Hoglan and Michael Packer. The Board has one seat open. If you wish to nominate yourself or another member please contact any board member. Following are a few more pics of the evening. Thanks to all of you for making SJAA a fun place to reach for the stars.



Constellation mugs were also retroactively handed out to SJAA's last year award recipients:

Left to Right: Jim Van Newland, Teri Kahl, Paul Mancuso, Bill O'Neil, Carl Reisinger

(Continued on page 3)

(Annual meeting and Potluck continued from page 2)

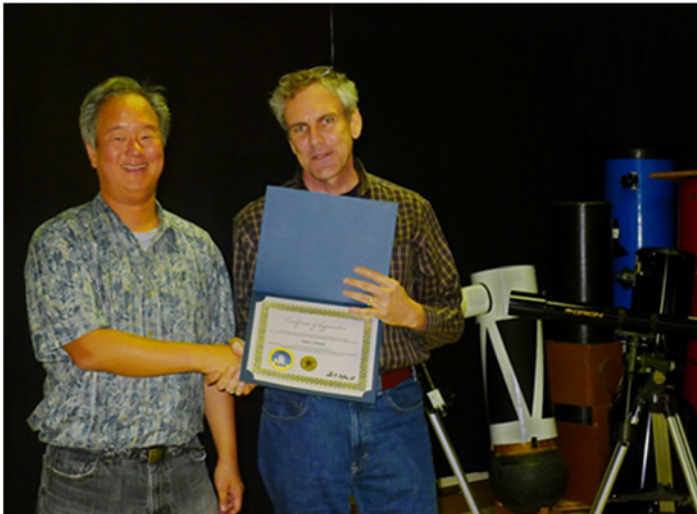
Below are the folks who were awarded for their SJAA efforts. Each recipient also got a bonafied astronomy cup that shows the constellations when it's heated up with coffee, tea, cocoa, broth... you get the idea.



Teri Rogoway given the SJAA Recognition Award by Greg Claytor for helping SJAA secure dark sky observing sites, late at night, and outside city lights. Much thanks Teri.



Greg Claytor and Teruo Utsumi receive awards on behalf of Marianne Damon and Tom Piller (not pictured). In addition to her outreach, Marianne was the reason every SJAA meeting had coffee and snacks last year. SJAA recognized Tom Piller for his outreach last year and his editorship of the SJAA Ephemeris News Letter. Thanks to you both!



Gary Chock has showed up and shared the night sky at just about every dark sky event SJAA hosted last year – Wow – Thanks Gary!



Below is a pic of three recipients in attendance might call these mug shots – seeing as they are holding their cups!
Pictured from left to right Gary Chock, Sukhada Palav, Teri Rogoway



Sukhada Palav has taken it upon herself to stock an Astronomy Library with books and magazines that both kids and adults can enjoy/utilize. Last year the Library really came to life!
Thanks Sukhada.

SJAA Member Observation Posts

Editor's Note: The following SJAA member posts came from the SJAA Group Yahoo page. I could not resist memorializing this email string about Sirius B in the Ephemeris. T. Piller

Sirius B 1/29/2015 From Mark McCarthy

One of the first books I read last year when I began to observe seriously (pun intended), was Burnham's *Celestial Handbook*. In it he describes Sirius and the history of the discovery of its companion, B, also known as "The Pup." Between 1834 and 1844 F.W. Bessel found irregularities in Sirius' motion through space, and calculated it must have a binary companion with a period of 50 years. But Sirius' brightness overwhelms the view of its companion. Alvan G. Clark was the first to make a visual observation, with his 18.5" refractor – then the largest in the world – in January 1862. Burnham then describes his own rapt experiences observing magnitude 8.75 Sirius B using various instruments ("very definite at 12 inches, difficult at 9 inches, and detectible at 6 inches only because its exact position was known" p. 395). Burnham then goes off on one of his wonderful tangents to discuss the nature of Sirius B, which is a white dwarf, with "a mass nearly equal to that of the Sun" packed into a star only the size of the earth, whose "luminosity, however, is less than 1/400 that of the Sun" (p. 397). Given that I was becoming very fond of observing double stars, I added Sirius B to my "bucket list" of objects to observe.

This observation may not be the monumental achievement it once was. Sirius B is now on its way toward maximum separation. It is currently 10.32 arc seconds, with position angle 078.5. Maximum separation is in 2023 at 11.28 arc seconds, PA 062.7. Nevertheless, it still generates excitement. Within the last couple of months, several local amateur astronomers have posted on internet forums that they had a clear split, one with a 16" and another with an 11" aperture reflector. The Double Star Observing section on the Cloudy Nights Forums

website has a pinned thread called "Da Pup is Busted," where many have posted their recent successful sightings, including photographs and field sketches showing its position relative to nearby stars. Surely there are many more people who've made sightings but didn't report them online. The most remarkable observation I've come across is that of Jay Reynolds Freeman, who split Sirius with a 6" f/8 refractor at Montebello Open Space Preserve in 2002, when the separation was a mere 5 arc seconds. The common requirement was very steady seeing conditions and generally high (250x or more) magnification. Aperture seems less of a determining factor, but the reports do follow Burnham's aperture-to-visibility scale as above. Steady seeing helps reduce the size of the roiling mess that Sirius presents in the eyepiece, providing the opportunity to see the comparatively faint B star just in or outside of A's diffraction pattern. Reading these reports gave me encouragement that I might be able to make the observation, but more importantly how I might.

I set my scope (12.5" f/7 reflector, 14.4% secondary central obstruction) in my Fremont, CA back yard – elevation 53' – upon returning home from work, at set a fan behind the primary to help it cool and track temperature. After dark I began observing various double stars and noticed I could push magnification rather high, 277x or more, and I was beginning to see nice Airy Disks. I paused at 7:30pm to attend to my evening chores, returning to the scope at 9:00pm. I noticed that Sirius was hardly twinkling – a sign that the seeing had become superb – and thought to make a go of it. I collimated my scope once more and put in a 9mm orthoscopic eyepiece (246x) which I had fitted with an occulting bar – a piece of black photographer's tape across the lens inside the barrel – which I planned to use to block Sirius A's light and which I carry around in my eyepiece bag for just such an occasion. Since Sirius B's PA is now 078, a little north of due east of Sirius, I let Sirius drift west through the field with the occulting bar set to the west. I tried several times, but no luck! I didn't

see anything but the roiling mess. I then used a 2x barlow with the same eyepiece (492x), and still did not see it. I thought maybe this wasn't the night. After all, there was a 73% waxing gibbous moon; SQM-L to the target was 17.52, and 18.22 at zenith. Transparency was probably average, and my scope was dripping dew.

Then I thought to use an intermediate magnification, and inserted a 5mm Nagler into the eyepiece into the focus, 443x, without an occulting bar. Sirius's diffraction was reduced versus the 9mm. I once again scanned the area east of the star. After three passes I noticed it: a faint yellow-orange disk, to the edge of Sirius A's diffraction and just below my spider's diffraction spike. I let Sirius pass beyond the field stop, and for an instant there was Sirius B, clearly separated, before it too passed beyond the field stop. I did it!

I kept looking at it for another 15 minutes, hardly able to contain my joy. One of the tips I had read in other observing reports is, if one is using a reflector, to rotate the tube assembly so the spider diffraction spike is not due east, which would interfere with the observation. I have no means to rotate my tube, so I decided re-observe in a couple hours, when the diurnal motion of the sky would have rotated the field for me. I observed a few more doubles in the meantime. Sigma Orionis was a special treat: the AB pair 0.25" separation appeared just barely, but clearly, notched. I came back to Sirius at 10:30pm, and sure enough B was halfway between my two spider diffraction spikes, plain as day. I kept observing it for another hour; I wanted to hold on to the magical feeling.

And so here I am this morning, not reading my work emails but typing up this report, elated. It was great. I hope to do it again.

Sirius B From Jack Zeiders

Wonderful stories! I'm quite pleased to
(Continued on page 5)

SJAA Member Observation Posts

(Continued from page 4)

see so many of the group doing doubles. It makes a nice change from the other various avenues of observing.

I first split Sirius back in the 80s with my then newly built 17" Dob at

Fremont Peak. If I recall I used a 7mm Meade research grade ortho, the 4mm MRG, plus a 8mm and 4mm Brandons. No occulting bar, just used the field stop trick to initially find it, After it was found direct vision worked as the seeing popped in and out. This was at full aperture, not using my sub-aperture mask. Due to the three vane spider, there are six diffraction spikes but they are half as bright and the tube of my Dob does not rotate.

I had tried previously with a ATM 8" f/8 and a 10" f/5 but had no luck.

The great thing about the "Peak" is the abundance of superb seeing nights. Even though the surrounding metropolitan areas have grown along with their attendant light pollution, the excellent seeing remains. Also as I was up there the night of the 23rd photographing comet Lovejoy and while the Hollister and Salinas sky glow is much larger and brighter than it was in the 60s, 70s, & 80s, it is still better IMO than the sites closer to the large metro area of San Jose. When the low summer marine layer comes in and blocks the urban lights it is like going back 40-50 years to the dark clear stable skies of yesteryear. A big plus is being at 2800 feet you are usually above the wet layer so it feels warmer and usually there is no dew so your gear stays dry. If one is willing to drive a bit, the chance of splitting Sirius is much greater there than on the valley floor. Either the SW lot or along the North West ridge area that over looks the bay and Santa Cruz are the best spots as you get first crack at the smooth laminar air off the Pacific.

I also purchased Burnham's handbook 3 volume set and read through them, Alone with the Webb series I think they are a wonderful resource for observers even today.

Thanks for sharing your story....Jack

Sirius B From Jim Van Nuland

Mark McCarthy wrote:

> Burnham then goes off on one of his wonderful tangents to discuss
> the nature of Sirius B, which is a white dwarf, with "a mass nearly
> equal to that of the Sun"....

Yes, Burnham is wonderful reading. I love his tangents. In the 1970s I made my way through all three volumes. Would that I could remember the half of it....

> I let Sirius pass beyond the field stop, and for an
> instant there was Sirius B, clearly separated, before
> it too passed beyond the field stop. I did it!

What a wonderful tale!

From my card file:

1976 Jan. 9 -- Using my 8 inch f/6 driven Newtonian at 305x, I saw the pup clearly by direct vision, during moments of best seeing.

1978 Jan.30 -- Suspected at 305x

1983, 84, 85, 87 -- not seen (none were excellent skies)

That is my most recent attempt. Your story reminds me to resume my efforts. Thank you!

> I kept looking at it for another 15 minutes, hardly able to contain my joy.

I can't read this without a big grin. Thanks yet again!....Jim

Short Test From Vini Carter

Pretty good backyard viewing right now. Got home at 11:00 pm after symphony rehearsal last night and the sky was clear. The new wooden "distressed" tripod from Oberwerk had arrived earlier and I wanted to see if it

made a difference over the Manfrotto tripod which I had robbed from my 4x5 camera. Wow! It wasn't just a little difference; there is about 10 times less vibration. (Thanks Ed for the tip about Oberwerk deals.)

So what I thought was going to be a short "test" lasted until 2:00 am. Jupiter was very high and I got some of the best views ever. (We began astronomizing in June 2014, so for Jupiter I guess that reads "in the last two months".) It was great to be able to focus and pan without waiting for the image to stop jiggling when using 250X.

Looked at Orion with the same high power and was able to see an almost three-dimensional "puffiness" to the nebula. Interestingly, I could easily split Mintaka with just 36X; however, the dim companion became difficult to see at 250X. Castor, on the other hand, required at least 150X to resolve into two stars with 90mm of aperture.

Then I put on a 36X eyepiece and just panned around the narrow bit of backyard sky available to me; found M36 just above the house. It didn't matter what lens I used; it was always very faint. With low power it appeared more like a tiny nebula, but with higher power, the stars began to be resolved. Tonight looks even better.

-21.23° S, 159.78° W, -10 GMT Rarotonga, Cook Islands

From Tom Piller

In November 2014 my wife Mary and I took a long planned trip to the Cook Islands to celebrate her 60th birthday. We flew Air New Zealand direct from LAX to the Cook Island called Rarotonga. Flight time was approximately twelve hours. We lodged in a place on the SSW side of the island. The little group of cabins where we stayed sat around the perimeter of a large grassy area that opened up to the ocean looking due south. We snorkeled during the day and when the skies were clear I was out looking at the southern stars by night.

We timed the trip to occur towards the end of the southern spring but just before monsoon season and of course the new moon.

(continued on page 6)

(Rarotonga Cont. from page 5)

Of the 12 nights we were there, 5 were good for star gazing; not bad for a tropical island in the middle of the Pacific. Below are excerpts from a log I kept of the trip.

Monday, 11/10/2014: arrived at the Rarotonga airport around 6am. Didn't do much except exchange for some local money, pick up groceries and do a little exploring. Cloudy all night.

Tuesday, 11/11/2014: Rained most of the day and hard throughout the night; good sleeping!

Wednesday, 11/12/2014: AM: went on a 2 hour snorkel swim; there are many more fish in this lagoon than I have ever seen in Hawaii. PM: cloudy until 9:30 then mostly clear. Got the binoculars out (8.6x56): and right away observed NGC 104 (47 Tucana; globular cluster) and the Small Magellanic Cloud (SMC) in the SW. Tried to locate the LMC but could not find it. Neither the LMC nor SMC were visible to my naked eye and just barely visible through the binoculars. Found Mars, saw the tail of Scorpius in the NW (it was upside down), Orion in the SE (Orion was also upside down), M45 in the East, and watched Canopus rise due south early evening and observed it on and off the entire night as it made a big loop around the SCP. Attempted taking some pictures but was not very successful due mostly to the wind. Called it quits around 11:30pm. Felt pretty good for what I was able to observe the 1st night out in unfamiliar territory. Need to figure out a way to cover up those dam landscape lights.

Thursday, 11/13/2015: AM: snorkeled for about 4 hours. 8PM: skies were partly cloudy and cleared up as the night progressed. Spent a fair amount of time getting to know some of the southern constellations and objects: HYDRUS, ERIDANUS/Achernar, RETICULUM, TUCANA/47 Tucana; GRUS/Gruid; INDUS; CORONA AUSTRALIS; SAGITARIUS (upside down); PEGASUS/M31 (Pegasus was right side up!). 9PM set up the AT72mm w/ 18mm eyepiece (24x) and spent more time observing 47 Tucana, along with M31, M42 and M45. Called it quits around midnight just as IC2604 (Southern Pleiades) was rising in the SSE.

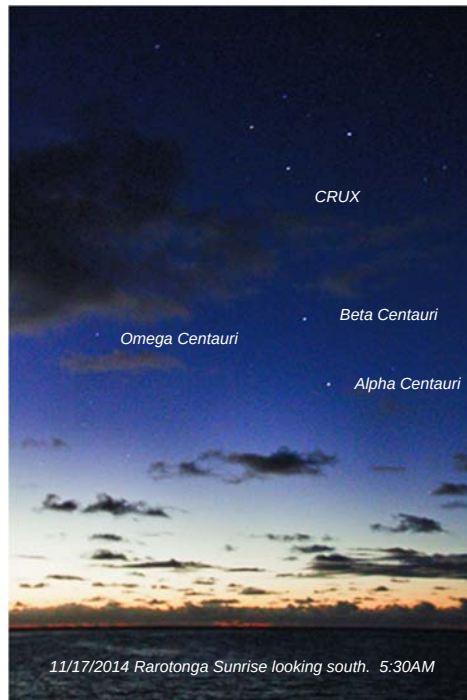
Friday, 11/14/2014 PM: went snorkeling, water visibility was not too good (10'). PM: overcast, no viewing.

Saturday, 11/15/2014: AM: water visibility not too good. Swimming back to shore at low tide we were attacked by trigger fish; creepy! PM: sky was overcast; no viewing.

Sunday, 11/16/2014: got up at 2AM to check the sky; cloudy, no stars out. AM: found a new way to wade out to the good snorkeling area and saw several coral heads and tons

of fish on the way out to the deeper part of the lagoon. PM: cloudy except for twilight when Mary and I sat out and watched the sun set and the first stars to come out; Achernar and Canopus.

Monday, 11/17/2014: 4AM: walked outside and was blown away; the sky was clear and dark and the Southern Cross was immediately obvious and really bright. Stacked from the horizon all the way up to the zenith were Alpha Centauri, Beta Centauri (Hadar), Southern Cross (CRUX), Southern Pleiades, Carina Nebula, Carina cross asterism, Canopus (Alpha Carinae). Wasn't able to see Omega Centauri (NGC5139) for some reason. I took every available seat cushion I could find and set out to cover up as many of the landscape lights as I could; worked pretty well. I set the camera up on the tripod with an 18- 200mm zoom. Took pictures of CRUX and as much of the southern sky as I could squeeze into the frame between the palm trees. Sat out on the deck and watched CRUX and Alpha Centauri fade away as the sun came up, snapping pictures and looking thru the binos. Called it quits at 6:30AM.



Tuesday, 11/18/2014: no viewing, cloudy

Wednesday, 11/19/2014: no viewing, cloudy. Traded emails with Ed Wong and Gary Chock. Ed wanted to know how the Carina group of objects looked and Gary was interested in the coal sack and the LMC and SMC.

Thursday, 11/20/2014: 3AM: intermittent clouds and rain drops but looking south and using the binos I was able to observe Omega Centauri, CRUX, NGC4755 (jewel box; just below CRUX), NCC4609 (Coal sack just below and right of CRUX); IC2944 Lambda Centauri cluster; IC2602 Southern Pleiades; NGC 3372 Carina nebula; NGC 3532 Open

cluster Carina; Avior, double star in the Carina cross asterism (couldn't split it with the binos); LMC (very faint but barely able to see NGC2070 Tarantula nebula. Turned in at 5AM.

Friday, 11/21/2014: 3AM: mostly cloudy but set up the AT72mm/18mm eye piece and in between clouds, I was able to catch glimpses of pretty much the same targets as the night before with binos except at 3 times the magnification. Spent a fair amount of time looking at Omega Centauri globular cluster at 24x, it is really big. Also homed in on the CRUX jewel box, NGC3766 (pearl cluster), NGC3532 (open cluster in Corina), NGC3372 'Keyhole nebula' or 'Carinae nebula' (was able to see dust lanes). This whole section of the sky (looking mostly south) was full of stars; wow.

NOON: last day of snorkeling. The trigger fish were really worked up today; on the way back from snorkeling as Mary and I were wading through the lagoon, one finally hit me and then Mary got bit; a nasty little bite. Well that's it for the snorkeling.

Saturday (last day), 11/22/2014: 3AM: sky was overcast and it started to rain so no stars to look at. Spent the day packing and getting ready to leave.

The plane leaves tonight at midnight; it will be good to get home.



Photo: Astro Tech 72mm mounted on a Televue Gibraltar Alt-Az head attached to a travel style tripod. Nothing over 18" long in this travel set-up.

SJAA's 3rd Year Solar Anniversary A Big Hit

Today with a Sunspot Count of 193 (NOAA) and one huge massive H-Alpha flare, SJAA shared some stellar views with the public along the Los Gatos Creek Trail at Campbell Park.



*Above Left Today's Sunspots and Active Regions and
Above Right Malika Carter Setting Up Her Eyepiece
Projection System*



Bill O'Neil with the Clubs H-Alpha OTA



Mark McCarthy's scope was a big hit



Justin Gallegos stopped by and snapped this iPhone 4s image of today's massive H-Alpha Flare. Thanks for sending our way! And below are more pics of today's event. All in all, we had about 60 folks stop by.

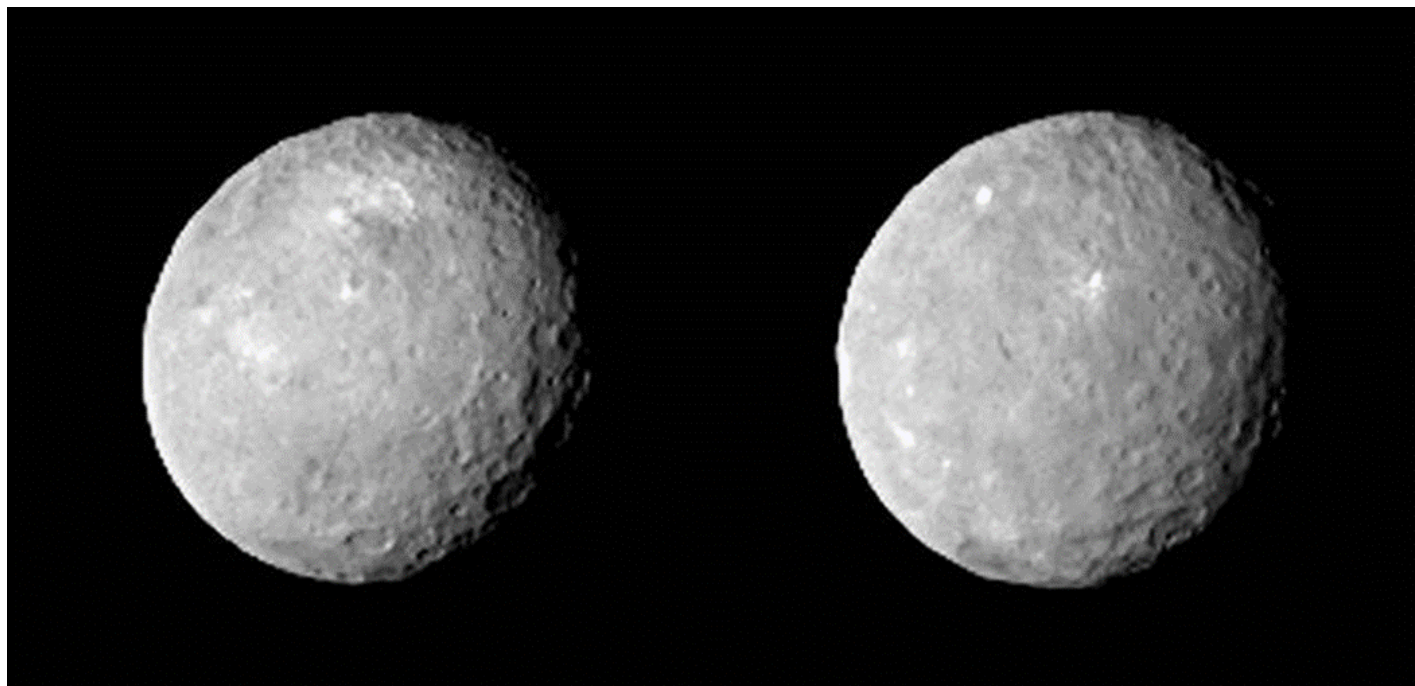


Veteran SJAA member and solar observer Terry Kahl at her grab and go Coronado H-Alpha. This scope revolutionized Solar Observing from an obscure hobby to popular astronomy



Bill O'Neil with a Mother and Daughter who stopped by.

Dawn Captures Sharper Images of Ceres



These two views of Ceres were acquired by NASA's Dawn spacecraft on Feb. 12, 2015, from a distance of about 52,000 miles (83,000 kilometers) as the dwarf planet rotated. The images have been magnified from their original size.

Image Credit: NASA/JPL-Caltech/UCLA/MPS/DLR/IDA

Craters and mysterious bright spots are beginning to pop out in the latest images of Ceres from NASA's Dawn spacecraft. These images, taken Feb. 12 at a distance of 52,000 miles (83,000 kilometers) from the dwarf planet, pose intriguing questions for the science team to explore as the spacecraft nears its destination.

"As we slowly approach the stage, our eyes transfixed on Ceres and her planetary dance, we find she has beguiled us but left us none the wiser," said Chris Russell, principal investigator of the Dawn mission, based at UCLA. "We expected to be surprised; we did not expect to be this puzzled."

Dawn will be gently captured into orbit around Ceres on March 6. As the spacecraft delivers better images and other data, the science team will be investigating the nature and composition of the dwarf planet, including the nature of the craters and bright spots that are coming into focus. The latest images, which have a resolution of 4.9 miles (7.8 kilometers) per pixel, represent the sharpest views of Ceres to date.

The spacecraft explored the giant asteroid Vesta for 14 months during 2011 and 2012. Scientists gained numerous insights about the geological history of this body and saw its cratered surface in fine detail. By comparing Vesta and Ceres, they will develop a better understanding of the formation of the solar system.

Credit: NASA

Exploded Star Blooms Like a Cosmic Flower



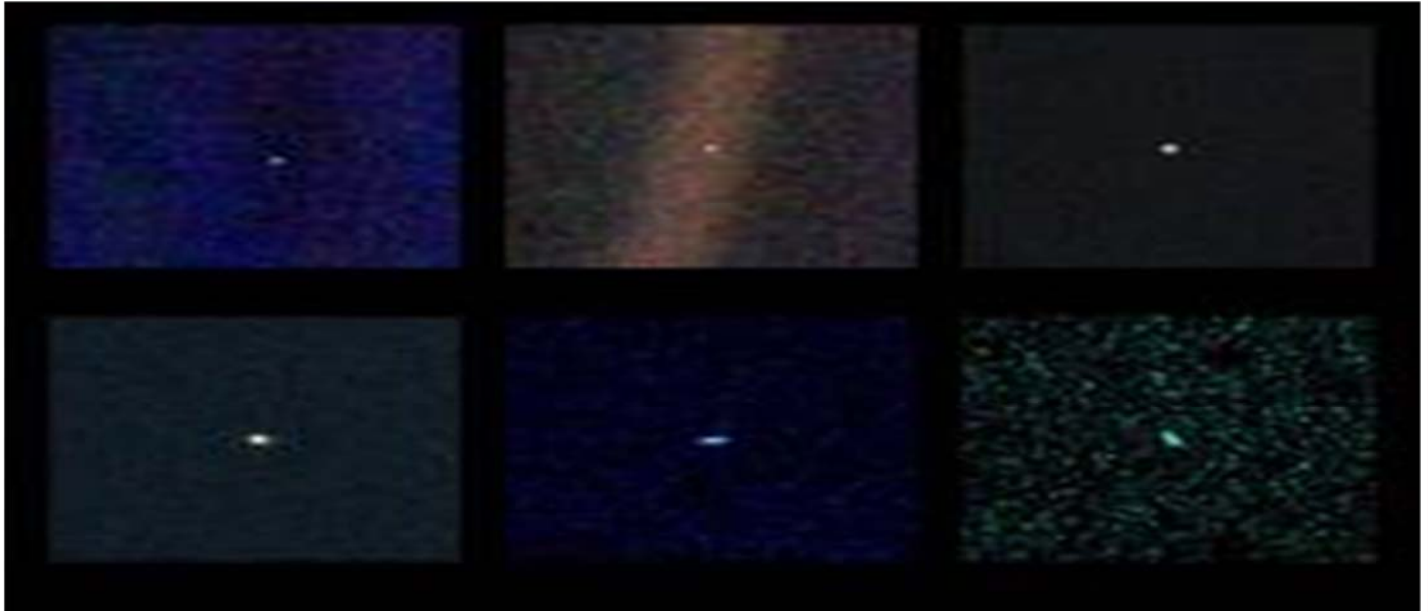
Chandra observations of the supernova remnant G299.2-2.9 reveal important information about this object.

Image Credit: NASA/CXC/U. Texas

Because the debris fields of exploded stars, known as supernova remnants, are very hot, energetic, and glow brightly in X-ray light, NASA's Chandra X-ray Observatory has proven to be a valuable tool in studying them. The supernova remnant called G299.2-2.9 (or G299 for short) is located within our Milky Way galaxy, but Chandra's new image of it is reminiscent of a beautiful flower here on Earth.

G299 was left over by a particular class of supernovas called Type Ia. Astronomers think that a Type Ia supernova is a thermonuclear explosion – involving the fusion of elements and release of vast amounts of energy – of a white dwarf star in a tight orbit with a companion star. If the white dwarf's partner is a typical, Sun-like star, the white dwarf can become unstable and explode as it draws material from its companion. Alternatively, the white dwarf is in orbit with another white dwarf, the two may merge and can trigger an explosion. Regardless of their triggering mechanism, Type Ia supernovas have long been known to be uniform in their extreme brightness, usually outshining the entire galaxy where they are found. This is important because scientists use these objects as cosmic mileposts, allowing them to accurately measure the distances of galaxies billions of light years away, and to determine the rate of expansion of the Universe. Credit: NASA

Valentine's Day Special for NASA's Voyager Mission



These six narrow-angle color images were made from the first ever "portrait" of the solar system taken by Voyager 1, which was more than 4 billion miles from Earth and about 32 degrees above the ecliptic.

Image Credit: NASA/JPL-Caltech

Valentine's Day is special for NASA's Voyager mission. It was on Feb. 14, 1990, that the Voyager 1 spacecraft looked back at our solar system and snapped the first-ever pictures of the planets from its perch at that time beyond Neptune. This "family portrait" captures Neptune, Uranus, Saturn, Jupiter, Earth and Venus from Voyager 1's unique vantage point. A few key members did not make it in: Mars had little sunlight, Mercury was too close to the sun, and dwarf planet Pluto turned out too dim.

Taking these images was not part of the original plan, but the late Carl Sagan, a member of the Voyager imaging team at the time, had the idea of pointing the spacecraft back toward its home for a last look. The title of his 1994 book, "Pale Blue Dot," refers to the image of Earth in this series.

"Twenty-five years ago, Voyager 1 looked back toward Earth and saw a 'pale blue dot,' " an image that continues to inspire wonderment about the spot we call home," said Ed Stone, project scientist for the Voyager mission, based at the California Institute of Technology, Pasadena.

The image of Earth contains scattered light that resembles a beam of sunlight, which is an artifact of the camera itself that makes the tiny Earth appear even more dramatic. Voyager 1 was 40 astronomical units from the sun at this moment. One astronomical unit is 93 million miles, or 150 million kilometers.

These family portrait images are the last that Voyager 1, which launched in 1977, returned to Earth. Mission specialists subsequently turned the camera off so that the computer controlling it could be repurposed. The spacecraft is still operating, but no longer has the capability to take images.

"After taking these images in 1990, we began our interstellar mission. We had no idea how long the spacecraft would last," Stone said.

Today, Voyager 1, at a distance of 130 astronomical units, is the farthest human-made object from Earth, and it still regularly communicates with our planet. In August 2012, the spacecraft entered interstellar space – the space between the stars – and has been delivering data about this uncharted territory ever since. Its twin, Voyager 2, also launched in 1977, is also journeying toward interstellar space.

Voyager 1 is more than three times farther from Earth than it was on Valentine's Day 25 years ago. Today, Earth would appear about 10 times dimmer from Voyager's vantage point.

Sagan wrote in his "Pale Blue Dot" book: "That's here. That's home. That's us. On it everyone you love, everyone you know, everyone you ever heard of, every human being who ever was, lived out their lives. ... There is perhaps no better demonstration of the folly of human conceits than this distant image of our tiny world."

Credit: NASA

SJAA Astrophoto Gallery



**IC1805 & IC1848 - Heart & Soul
Nebula
From Ed Wong**

*Location: RCDO
Camera: Full Spectrum Canon T3i
Imaging Lens: Nikon 70-200mm F2.8
Mount: Orion Sirius Mount
Auto Guider: Orion Mini Guider
Finder Scope: Orion Short Tube 80
Notes: Camera and lens was piggy-backed onto Orion Short Tube 80 in order to find object. Actual image was taken through the Nikon Lens which give a field of view of 6.58 x 4.58 degrees
ISO 1600 @ F2.8
Light Frames 30x 150 sec. exposures
Bias Frames 23x 1/4000 sec. exposures
Flat Frames 26x 1/200 sec. exposures
Dark Frames 20 x 150 sec. exposures
Processed with Deep Sky Stacker, Photoshop, Nikon Capture*

**Comet Lovejoy 01-13-2015
From Jack Zeiders**

Nikon D800 30s @ f/2.8 Nikon 300mm f/2.8 EDIF ISO 3200 taken near Fremont Peak, CA cropped from a single full frame

I went up to the Peak last night for another attempt to photograph the nice big comet we currently have gracing our night sky. It was not as dark as I had hoped due to a lack of fog cover for either Hollister or Salinas. It was a bit cold with a light breeze. I only shot a few single frame pictures at High ISO and a fast lens to attempt to cut through the light pollution and capture as much of the comet as I could.

So the picture is noisier than I would like.

(Editors Note: Jack's reference to 'the Peak' is in reference to Fremont Peak)



**The Cone, NGC 2264
From Paul Mahany**

Taken with a Canon 60Da with a Paracorr on an ATM 6" F5 Newtonian on a EQ6 Mount. Controlled with Sky Tools Pro. Image acquisition by BYEos. Guided with a 50mm PHD Guider.

93@ 420s @ iso 800 over 4 nights ending 1/23.

Stacked with DSS with flats, bias, darks.

Processed with StarTools.

SJAA Astrophoto Gallery



M81-M82
From Alex Kramarov

*The first image that i am actually proud of, M81-M82. Taken at Henry Coe overflow lot on Jan 24th, 12*300sec subs + 10 darks, Celestron 80ED (600mm F7.5), Canon t4i unmodified, IDAS LPS-D1, ISO 800, stacked with Deep Sky Stacker, curves and levels tuning with Photoshop*

Comet C/2014 Q2 Lovejoy
From Rogelio Bernal Andreo

This thing (Comet C/2014 Q2 Lovejoy) isn't just bright. The tail is huge!!!! There's like 5+ degrees there and I'm sure it can go all the way to 7+

I captured this photo Tuesday night from Montebello OSP. The sky was rather bright (mag 19.5 at best). I also improvised a bit, and was constantly "distracted" by the gang that had gathered up there for the occasion (not to blame them, I am the one who gets easily "distracted" ;-). I can only imagine going for it under nice dark skies and good conditions. Phew....



Rogelio Bernal Andreo
DeepSkyColors.com



Solar Eclipse 10/23/2014
From Aaron Turner

This is a photo of the solar eclipse I took back on 10/23/14 at 2:43:42 pm in Sunnyvale.

I used my iPhone to take the photo through my PST telescope. No post processing or anything like that.

SJAA Astrophoto Gallery



Orion Nebula (above)
From David Rogers

*Taken on Summit Rd just west of HWY 17
Orion XT8 Classic (basic dob mount) using prime focus
on a Sony NEX-6
f/5.9, ISO 25,600, 0.4 second exposures
Best 80 of 160 light exposures, no darks, bias, or flats
used
Stacked with DSS and post-processed with Lightroom*

Southern Sights

From Tom Piller

Location: Rarotonga, Cook Islands (looking South)

*Photo above: CRUX (bottom) Carina (center to top)
Canon EOS60D, 20mm, ISO 3200, 10 sec, no stack*

*Photo to the Right: Constellation Orion; upside down
Canon60D, 32mm, ISO 1600, 20 sec, no stack*



Winter Deep Sky Scavenger Hunt (DSSH)

From Ed Wong

Thanks to everybody who came out for the February 14th event. I know it was hard trying to complete the event after a month went by but, it's hard to predict the weather. I really appreciate those who were able to make it out. We had a smaller group last night, about 10 of us mostly due to Valentine's Day plans and a 3-day weekend for some people. We had 3 couples who did celebrate Valentines Day "under the stars" :-)

The event started off at 7pm sharp, the weather was clear the whole evening and ended up not being too cold which was nice. The seeing was not too bad, there was some moisture in the air that kept the skies from being as dark as it can be at MR. The best SQM reading I was able to get was 20.65.

The event was very fun for me personally, nice to see everybody going through the list and learning to find new objects. I went through then entire intermediate list with my binoculars but, could not find NGC404 (Mirach's Ghost) it's a pretty tough object for the smaller scopes. If you haven't looked for it before it's a good object to try find.

It was also nice to have my wife Shelley come out and view with me. It was her first time out this winter viewing. Other couples were Paul & Marion and Patrick & Emmanuelle.

Below I have listed the results of the trophy dash and bragging rights until the next Scavenger Hunt:

Beginner's List : Shelley Ahn-Wong found all 30 objects in about 2 hours using a 4 inch refractor and a 10x42mm binocular.

Intermediate List: Patrick & Emmanuelle Debaumarche found all 30 objects in about 3 hours using a 16 inch dob and a 80mm retractor.

Advanced List: Gary Chock found 22 of the 30 objects using a 10 inch dob and a 28x110mm binocular

Congratulations to the new champs!!!

For those who were not able to attend, you can still try out the lists. Most of the objects should be up for next few months. Starry Nights next weekend would be a good chance to show off some of the new objects.

Looking forward to seeing everybody out next time under the stars.

From Gary Chock: Thanks to Ed for the challenge! And to Teruo for the competition in the Advanced List which helped to motivate me. It was great to see everyone at dusk while setting up. Now, onward to what I couldn't find because of the conditions and to the Bonus List!

This New Moon marks the Lunar New Year. Happy Year of the Sheep / Goat / Ram 2015!

From Emmanuelle: Ed, Thanks so much for making this wonderful event happen! We had a fantastic time, and a trophy to show off :) Next time we are going to try locate the objects from the advanced list!





Kid Spot Jokes:

- ♦ **What hot drink do aliens enjoy?**
Gravitea
- ♦ **What is a light year?**
The same as a regular year but with less calories.

Kid Spot Quiz:

- A. Which planet has winds blowing up to 1,240 miles per hour?
- B. Which planet has the lowest density in the solar system?

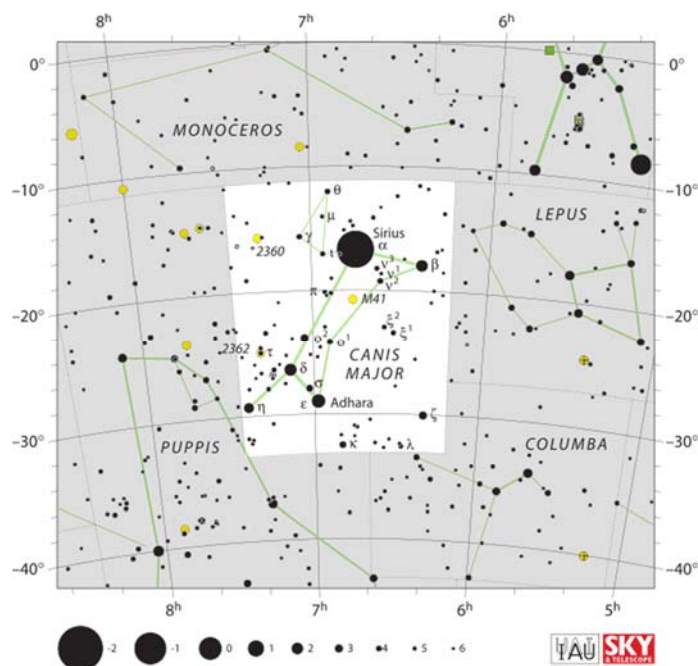
Kid Spot Space Word Challenge: Have fun unscrambling these simple words.

1. NOMO
2. RUNAST
3. BTIOR



Image credit: NASA/JPL-Caltech/
Space Science Institute

Constellations



Canis Major

Canis Major is a constellation in the southern celestial hemisphere. In the second century, it was included in Ptolemy's 48 constellations, and is counted among the 88 modern constellations. Its name is Latin for "greater dog" in contrast to Canis Minor, the "lesser dog"; both figures are commonly represented as following the constellation of Orion the hunter through

the sky. The Milky Way passes through Canis Major and several open clusters lie within its borders, most notably M41.

Canis Major contains Sirius, the brightest star in the night sky, known as the "dog star". It is bright because of its proximity to our Solar System. In contrast, the other bright stars of the constellation are stars of great distance and high luminosity.

In Greek Mythology, Canis Major represented the dog Laelaps, a gift from Zeus to Europa; or sometimes the hound of Procris, Diana's nymph; or the one given by Aurora to Cephalus, so famed for its speed that Zeus elevated it to the sky. It was also considered to represent one of Orion's hunting dogs, pursuing Lepus the Hare or helping Orion fight Taurus the Bull; and is referred to in this way by Aratos, Homer and Hesiod. The ancient Greeks refer only to one dog, but by Roman times, Canis Minor appears as Orion's second dog.

Credit:
Wikipedia

Kid Spot Quiz Answers:

- A. Neptune
- B. Saturn

1. MOON
2. SATURN
3. ORBIT

SJAA Library!



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astron-

mers!

Please check out our wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

Thanks to Dave Ittner, Teruo Utsumi and Vini Carter...the SJAA library is now visible and accessible!

Recently 3 books were added:

- Stargazing for Dummies
- Objects in the Heavens
- Double Stars for Small Telescopes: More Than 2,100 Stellar Gems for Backyard Observers

One of the members checked out the double stars book as soon as I officially added it to the library, during the Fix-It on 2/1!

Request to all - Please check with club Librarian Sukhada Palav before accepting magazine donations for the library. It makes sense not to add more at this point as we don't have any room left to store them.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

Weather doesn't slow this program down, so even if it's windy, cold and rainy outside, Fix It day goes on. Big thanks to go Ed Wong, Phil Chambers,

and Dave Ittner for being the gear experts who faithfully make themselves available at Fix It day along with Vini Carter who has stepped up to take over as Fix It program chair from Ed Wong.

Solar Observing

Solar observing sessions, headed up by Michael Packer, are held the 1st Sunday of every Month from 2:00-4:00 PM at Hogue Park weather permitting. See article and photos on page 7.

Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Hogue Park. This is a regular, monthly session, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects. We'll need extra help to pull this off, but it appears to be doable."

The announcement and background can be found here:

<https://groups.google.com/forum/#!topic/sjaa-announcements/ScBqsdpmzq>

Loaner Program

The purpose of this program, headed up by Manoj Koushik, is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment below. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been

through the SJAA Quick STARt program please fill this form to request a particular item. Please also consider donating unused equipment.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.



Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.

SJAA Contacts

President:	Rob Jaworski
Vice President:	Lee Hoglan
Treasurer:	Michael Packer
Secretary:	Teruo Utsumi
Director:	Greg Claytor
Director:	Dave Ittner
Director:	Ed Wong
Director:	Bill O'Neil
Director:	open
Memberships:	Ed Wong
Beginner Class:	pending
Fix-it Program:	Vini Carter
Imaging SIG:	Mark Striebeck
Library:	Sukhada Palav
Loaner Program:	Manoj Koushik
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Publicity:	Rob Jaworski
Questions:	Lee Hoglan
Quick STARt	Dave Ittner
Solar:	Michael Packer
School Events:	Jim Van Nuland
Speakers:	Teruo Utsumi
E-mails:	

<http://www.sjaa.net/contact>

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Club Updates

From the Board of Directors

Announcements

Annual Board elections were held at the Membership Meeting February 2015. The following Board members were up for re-election: Lee Hoglan, Michael Packer, Teruo Utsumi, Rich Neuschaefer and Ed Wong. With the exception of Rich Neuschaefer, who had decided to not seek re-election, the above board members were all re-elected. With Rich's departure, one board seat remains open. See Feb 07, 2015 Board meeting excerpts on this page for additional information.

The board will vote on officers at the March 7, 2015 board meeting. There will be no General Meeting after the Board meeting.

The Half Messier Marathon will be held at Mendoza Ranch on Saturday, March 21.

The SJAA annual auction will be held on Sunday, March 8th. See SJAA web page for start - close times.

Board Meeting Excerpts

December 6, 2014

In attendance

In attendance: Rob Jaworski, Lee Hoglan, Dave Ittner, Bill O'Neil, Michael Packer, Greg Claytor Teruo Utsumi
Absent (excused): Ed Wong, Rich Neuschaefer
Guests: Vini Carter, Jordan Makower, Gary Kettleson, Ken, Chuck Pooley, Mural Walker

Mount Umunhum Project

There are signs of life for the concrete block. Mural Walker described a proposal to use the existing tower building for a publicly accessible observatory. Rob will be main point of contact for investigations into viewing/seeing conditions and plans to contact Mt Umunhum conservation Project about access to the site. Mural will further explore the cooperative effort required to move forward and later form a committee.

Astronomes sans Frontières

Rob Jaworski learned via an ASP conference call that AWB's Global Pairing Program may provide an interesting opportunity for SJAA to partner with international sister clubs. Michael requested 3 club names for the list. Rob will post AWB links to the board list, see what additional information is available and put the list out to public announcement.

January 3, 2015

In attendance

Rob Jaworski, Lee Hoglan, Dave Ittner, Bill O'Neil, Michael Packer, Teruo Utsumi
Absent (unexcused): Greg Claytor
Guest: Vini Carter, Tom Piller

Overnight Star Party in 2015

Ed has been looking at Pinnacles (east side), Lake San Antonio for an overnight star party. Both sites were discussed. The Pinnacles group sites allow 5 cars per site. Some concern was expressed concerns of possible public interference since the group sites are not isolated. Everyone expressed support for the idea. Ed will move forward to reserve Pinnacles for two nights.

Ephemeris Printed Version

Whether or not to continue the print version of the Ephemeris was discussed. The outcome of the discussion was that there are no plans to discontinue the printed edition.

Auction

The Club auction is coming up soon in early March. Dave Ittner volunteered to lead the Club's auction effort this year.

Volunteers

Teruo asked everyone to be mindful of people who can potentially help at various Houge events. Several people previously signed up on the Google form. Teruo to forward sign ups to program chairs.

Beginners Class Program Futures

Teruo reported that Bill, Susan O'Neil, Dave Grover, Dave Ittner, Rob Jaworski, Marianne Damon, and Teruo met Jan. 3, 2015 to discuss the future of the Beginners Class. Dave G. expressed his observation that the classes should target younger children.

February 7, 2015

In attendance

In attendance: Lee Hoglan, Bill O'Neil, Michael Packer, Greg Claytor, Rich Neuschaefer Teruo Utsumi
Absent (excused): Rob Jaworski, Dave Ittner
Guest: Vini Carter

Solar program

Michael Packer requested a budget for and was approved to purchase drinks for 2015 solar observing events.

Open board seat

The anticipated board seat opening (since Rich will not seek re-election) was discussed. There is no immediate need to fill the open seat. Ed talked w/ several people. The following names were forwarded as possible candidates: Malika Carter, Marianne Damon, Vini Carter, Sukhada Palav. Board members will approach the above people about filling the vacancy. The board would like to thank Rich for his service to the club over the years ... and congratulates him on his engagement!

Officer Election

Rob Jaworski's two year term as president ends as of the March 2015 board meeting. The board will vote on officers at the March board meeting.

Half Messier Marathon

To be held Saturday, March 21 at Mendoza Ranch, Teruo Utsumi will organize the event.

SJAA Forum Status

About 20 have been using the forum as a trial. Further details to be provided by Dave Ittner at next board meeting.

In Memory of Robert Armstrong

There are several possibilities of memorializing Robert, including naming the solar scope after him. Greg proposed creating an annual award: "Robert Armstrong President Award" to be decided by the president and awarded to an SJAA member for contributions to the club. Greg will draft a proposal for next month's board meeting.

Place
postage
here

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

Membership Type:

- ☐ Regular — \$20
☐ Regular with Sky & Telescope — \$53
☐ Junior (under 18) — \$10
☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2-3 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at:**

<http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to
sjaamemberships@gmail.com

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at <http://www.sjaa.net/join-the-sjaa/>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____